

ANDREAS TARASIDIS

Curriculum Vitae

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📍 Volos, Greece

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EDUCATION

Diploma (M.Eng) in Computer Science and Engineering University of Ioannina

📅 07/2025

Diploma Thesis

- "Deep Illumination-Driven Light Probe Placement"

This work presents an AI-based method for optimizing real-time global illumination by predicting high-importance light-probe locations with a PointNet-derived Light-Probe Neural Network (LPNN). The model analyzes voxelized lighting features such as RGB and normal variance, occlusion factor, and spherical harmonic values to assign an importance score to each grid position. Supervised labels are generated from existing optimized layouts, enabling the network to learn effective placement strategies. By combining local and global context through convolution and pooling, the approach produces efficient probe layouts that can be refined in Unity. The method reduces probe count and runtime cost while preserving lighting quality and generalization.

Relevant courses

- Software Development
- Computational Mathematics
- Artificial Intelligence
- Computer Graphics

PROJECTS

Robotics

Designed a system for calculating the trajectory of the wheeled robot TurtleBot3 in a ROS simulated environment using C++ and Python, as well as a real robot in the lab, using the Linear functions with parabolic blends method.

3D Computer Graphics

- Developed applications in pure C++ and OpenGL able to capture user input to change the environment in a 3D environment, with sounds, movement and simulation pausing.
- Developing a 3D video game from the ground up using Rust and Vulkan, with fully integrated modding systems and high-performance architecture.
- 3D video games made in Godot engine using C#, with focus on replayability, visual style, and multiplayer systems.

Software Development

- A CLI application made in Python that is capable of turning any image into ASCII art. Allows for various options.
- Creating an application using pure Rust that packs various useful utilities into a single application. Main focus is ease-of-use as well as very low resource requirements.

ADDITIONAL ACTIVITIES



Compulsory Military Service 2025 - 2026

Promoted to a Green Beret in the Greek Special Forces.



Eurographics 2026

Published Work as a Poster

INTERESTS

Technology

Software

Computer Graphics

Learning

Driving

Nature

Art

LANGUAGES

English



German



Greek



Norwegian



SKILLS

Engineering

Capable of combining scientific principles and tools from different fields while aiming at the completion of the task at hand.

Programming

- Rust, Python, C#, C, C++, Java, Assembly, VHDL, Verilog, Matlab
- Software life-cycle and Architecture
- Linux, Command-Line Interfaces, Computer Graphics, OpenGL, Vulkan
- Unity, Unreal Engine, Godot

Hardware

- VLSI Circuits, Electronics, Digital Circuit Design, Microprocessors
- Robotics

Soft Skills

- Teamwork, Dependable, Precise, Optimist, Problem solver

SEMINARS

12th Greek Electrical and Computer Engineering Students Conference, Greece.